AN INDEPENDENT EXACT VERIFICATION OF THE VARIATIONAL STEP BEHIND THE CLAIMED 2026 BOUNDS $H_1 \le 240$ AND $H_1 \le 236$, WITH A REMARK ON THE CAPS OF A 212 DRAFT

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ABSTRACT. Several 2026 preprints claim improvements of the Polymath8b bound $H_1 \le 246$. Each combines equidistribution estimates and a reduction theorem with a *variational step*: a test function on a capped region whose sieve ratio exceeds a threshold (4 in the Polymath scaling). We recompute the variational step of three claims in exact rational arithmetic, each capped ratio by two independent programs, and do not address the analytic input. (i) Stadlmann's support ($k = 49$, claimed 240): our test function gives $4.0120389\ldots$, and $4.0124198\ldots$ after optimising on the capped region. (ii) The support of Kintali and of Song–Yue ($k = 48$, claimed 236): for the published witness of Song and Yue, the cap-free mass and the capped numerator equal their reference integers (655 and 715 digits) exactly, so their criterion $48J/I_0 > 2001/2000$ holds, by a margin of $1.6 \cdot 10^{-4}$; the capped ratio $4.0030988\ldots$ agrees with all 33 digits printed by Kintali; our own test function gives $4.0060814\ldots$. (iii) The draft of Charton et al. ($k = 45$, claimed 212): the criterion holds for the caps of its Table 3 ($4.0027351\ldots$), but for the caps of its Lemma 9.1 no polynomial in the authors' own 846-dimensional space reaches 3.99 (ball arithmetic, from one program's Gram weights). This locates an inconsistency in the draft, not an error in its result. None of this verifies the analytic input, and none of it bears on the twin prime conjecture, which remains open.

1. INTRODUCTION

The reduction. For an admissible k-tuple $\mathcal{H} = \{h_1 < \cdots < h_k\}$, let DHL$[k, 2]$ denote the statement that infinitely many translates $n + \mathcal{H}$ contain at least two primes. If DHL$[k, 2]$ holds and $\mathcal{H}$ is an admissible k-tuple of diameter D, then $H_1 \le D$. Maynard's method [4] proves DHL$[k, 2]$ from a level of distribution and a symmetric test function F whose ratio $kJ(F)/I(F)$ exceeds a threshold; Polymath8b [5] enlarged the region on which F may live and obtained $H_1 \le 246$ with $k = 50$. The 2026 preprints listed below enlarge the region further, to *capped* supports, and pay for this with equidistribution estimates for the moduli that arise: Stadlmann [7] proves new ones, Kintali [3] uses Stadlmann's [7, Prop. 3] with one branch made empty, Charton et al. [1] use Stadlmann's Type I, II and III estimates for their own family of moduli, and Song and Yue [6] prove their own.

The claims examined.

claim	k	source	what we check
$H_1 \le 240$	49	Stadlmann [7]	variational criterion
$H_1 \le 236$	48	Kintali [3]; Song–Yue [6]	variational criterion; published witness
$H_1 \le 212$	45	Charton et al. [1]	variational criterion; caps

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Prepared with the assistance of Claude (Anthropic): the programs, the Lean files, the text and the internal referee reports were written by AI agents under the author's direction, and "independent programs" below means programs written in separate agent contexts without access to each other's code. All programs, certificates and Lean sources are in the repository https://github.com/jerryckim16/twin-primes; a living companion paper with full details is at https://jerryckim16.github.io/twin-primes/.

Each claim has three parts: an *analytic* part (an equidistribution theorem for the relevant family of moduli), a *reduction* from the variational criterion to DHL$[k, 2]$ ([7, Prop. 1], [6, §§4.12–4.14], [1, Thm. 4.10] and Kintali's use of Stadlmann's), and a *variational* part (a function F on the capped support with ratio above the threshold). We check only the variational part, and the combinatorial fact that an admissible tuple of the stated diameter exists. Throughout we label statements Computed (a machine computation, in exact rational arithmetic or in rigorous ball or interval arithmetic as stated, with a script that reproduces it; "two programs" means two independent programs agree exactly), Proved here, Lean-checked, Literature (read in the source), or Claimed (asserted in a preprint and not verified by us).

Results in brief. Ratios in the Polymath scaling, where the threshold is 4:

claim	variational ratio	status here
240	4.0120389; optimised on T: 4.0124198	Computed, two programs
236, witness of [6]	$48J_T/I_T = 4.0030988$; $48J_T/I_0 = 4.0026597$ $(= 4 \times 1.00066492$; SY criterion $> 2001/2000)$	Computed, two programs (J_T; I_T one)
236, our own F	4.0060814	Computed, two programs
212, Table 3 caps	4.0027351	Computed, two programs
212, Lemma 9.1 caps	< 3.99 for all F in their 846-dim. space	Computed, ball arithm., one program's weights

What this note does not do. It does not verify the equidistribution estimates of [7], [3], [6] or [1], nor their reduction theorems; all three claimed bounds remain conditional on them, and the established bound is still 246. Song and Yue [6, Remark 4.23] point to a missing hypothesis in one step of the proof of [7, Lemma 5]; we read that step and comment on it, and on which of the claims may depend on it, in Remark 10, without resolving it. Nothing here concerns the parity barrier or the twin prime conjecture.

2. The variational problem on a capped support

Normalisations. The sources work in coordinates t in which the criterion reads $kJ/I > 1$ (Song and Yue use the slightly stronger $48J/I_0 > 2001/2000$, see below). We use the Polymath scaling $u = 4t$, in which the criterion reads $kJ/I > 4$: every length is multiplied by 4 (so $g = 4\delta$ and $C_m = 4B_m$ for the threshold δ and caps B_m of [7], and the radius $U_{\mathrm{SY}} = 521/2000$ of [6] becomes $L = 1.042$), and every ratio kJ/I is multiplied by 4. Values quoted from the sources have been converted, except where we say otherwise.

Supports. Let $|u| = u_1 + \cdots + u_k$, $\mathcal{R}_k = \{u \in [0,1]^k : \ |u| \le 1\}$, and write $L\mathcal{R}_k$ for its dilate. The one-band supports of [7] (Literature: Definitions 1 and 5 and Proposition 1 of [7]) are described by a length L, a marginal radius τ, a threshold g and caps $C_1 \le C_2 \le \dots$, with $C_0 = 0$; with $\mathcal{L}(u) = \{i : u_i > g\}$ (the *large set*), $m(u) = |\mathcal{L}(u)|$ and $s(u) = \sum_{i \in \mathcal{L}(u)} u_i$,

$$T = \{u \in L\mathcal{R}_k : \ s(u) \le C_{m(u)}\}.$$

For a symmetric F supported on T,

$$I_T(F) = \int_T F^2, \qquad J_T(F) = \int_{\tau\mathcal{R}_{k-1}} \left(\int F(u', u_k)\, du_k \right)^2 du',$$

and the variational criterion is $kJ_T(F)/I_T(F) > 4$. (This one-band form of J_T, with marginals over $\tau\mathcal{R}_{k-1}$, is our transcription of the criterion of [7, Prop. 1]; see [2, §5].) We write I_0, J_0 for the same functionals without caps, that is with T replaced by $L\mathcal{R}_k$. In this scaling the parameters are as follows (Literature, converted).

	k	(L,τ)	g	caps $C_1, C_2, \dots$
[7]	49	(1.042, 0.982)	0.112	0.6, 0.6, 0.68, 0.68, …
[3], [6]	48	(1.042, 0.982)	91/1250	1899/2500 for all m
[1], Tab. 3	45	(53/50, 249/250)	41/625	$4B_{1,m}$ (see below)
[1], L. 9.1	45	(53/50, 249/250)	41/625	0.62, 0.62, 0.68, 0.68, …

For [1], Table 3 lists $B_{1,m} = \frac{1}{5000}(777, 794, 875, 917, 953, 983, 1016, 1042, 1063, 1081)$ for $m \le 10$ and $B_{1,m} = 1081/5000$ for $m \ge 11$, and $C_m = 4B_{1,m}$; Lemma 9.1 states $C_1 = C_2 = 31/50$ and $C_m = 17/25$ for $m \ge 3$. All four cap sets satisfy $g < C_m \le C_{m+1} \le C_m + g$ for $m \ge 1$ (checked by hand and in Python; the case of the Table 3 caps is also LEAN-CHECKED, `charton_caps_valid`). The inequality fails at $m = 0$ for every cap set, which is why Proposition 2 assumes it only for $m \ge 1$.

Exact moments. All integrals reduce to moments over simplices.

Lemma 1 (PROVED HERE; elementary)**.** *Let $P_j = \sum_i u_i^j$, $X_1, \dots, X_k$ independent exponential variables of mean 1, and $E_k(b) = \mathbb{E}\prod_j P_j(X)^{b_j}$. Then, with $w(b) = \sum_j j b_j$,*

$$\int_{\mathcal{R}_k} (1-P_1)^A \prod_j P_j^{b_j}\, du = \frac{A!}{(k+A+w(b))!}\, E_k(b), \qquad \sum_b E_k(b)\frac{s^b}{b!} = \left(\sum_c w(c)!\,\frac{s^c}{c!}\right)^k.$$

Proof. The Dirichlet integral gives $\int_{\mathcal{R}_k}(1-P_1)^A u^\alpha\, du = A!\,\alpha!/(k+A+|\alpha|)!$ and $\alpha! = \mathbb{E}X^\alpha$; sum over the monomials of $\prod_j P_j^{b_j}$, all of degree $w(b)$. The generating function follows from independence. □

The fibres of a capped support. Let $R = L - |u'|$ on the fibre over $u' \in \mathbb{R}^{k-1}_{\ge 0}$, and call u' *admissible* if $s(u') \le C_{m(u')}$.

Proposition 2 (PROVED HERE; elementary)**.** *Let $C_0 = 0$ and suppose $C_{m+1} \le C_m + g$ for all $m \ge 1$. Then the fibre $\{t \ge 0 : (u', t) \in T\}$ is empty when u' is not admissible and equals $[0, \theta(u')]$ otherwise, where*

$$\theta(u') = \min\big(R,\ \max(g,\ C_{m(u')+1} - s(u'))\big).$$

Consequently, with $Q(u', x) = \int_0^x P(u', t)\, dt$ for a polynomial P,

$$J_T(P\,1_T) = J_0(P) - \int_{\substack{u' \text{ not adm.} \\ |u'|\le\tau}} Q(u', R)^2 - \int_{\substack{u' \text{ adm., } \theta<R \\ |u'|\le\tau}} \big(Q(u', R)^2 - Q(u', \theta)^2\big),$$

and $I_T(P\,1_T) = I_0(P) - \int_{L\mathcal{R}_k \setminus T} P^2$.

Proof. If $t \le g$ the large set of (u', t) is $\mathcal{L}(u')$, so $(u', t) \in T$ if and only if u' is admissible and $t \le R$. If $t > g$ the large set gains the last coordinate and the condition is $s(u') + t \le C_{m(u')+1}$. If $m(u') = 0$ then $s(u') = 0 = C_0$, so u' is admissible. If $m(u') \ge 1$, the hypothesis $C_{m+1} \le C_m + g$ with $t > g$ forces $s(u') < C_{m(u')}$. So in either case a non-admissible u' has an empty fibre, and for admissible u' the fibre is $[0, \min(R, g)] \cup (g, \min(R, C_{m+1} - s)] = [0, \theta]$. The marginal of $P\,1_T$ is $Q(u', \theta)$ on admissible u' and 0 otherwise; subtract from the cap-free marginal $Q(u', R)$. □

By symmetry each correction is a finite signed sum, over the size m of the large set and over $j \ge 0$ further coordinates forced above g (inclusion and exclusion for "all other coordinates are at most g"), of integrals of polynomials over polygons in two radial variables; so every term is an exact rational number. The ranges that occur are short (PROVED HERE, elementary: m coordinates above g have sum $> mg$). The integer arithmetic behind each bound is checked

in Lean, in units of 1/1000, 1/1250 and 1/2500 (file `lean/TwinPrimes/Caps.lean`; the Lean statements are about lists of natural numbers, not real coordinates): at most 9 coordinates of a point of $L\mathcal{R}_{49}$ exceed g for [7] (`stadlmann_large_count`), at most 10 large coordinates of a point of T for [3] (`kintali_large_count`), and at most 16 coordinates of a point of $L\mathcal{R}_{45}$ exceed g for [1] (`charton_large_count`); and the inclusion–exclusion over further coordinates stops at $j \le 3$, 3, 6 respectively (`stadlmann_extra_le_three`, `kintali_extra_le_three`, `charton_extra_le_six`; the last assumes the large set carries at least 777/1250, so it covers the caps of Table 3 but not those of Lemma 9.1, for which only `capped_check.py` is used). The correction pieces themselves include non-admissible bases, whose size m is bounded only by the total length (up to about 14 for [3]). The reduction to two radial variables is written out in [2, §5].

3. Method

Test spaces. For $D \ge 0$ let V_D be the space of symmetric polynomials spanned by $(L - P_1)^a \prod_{i\ge1} P_{2i}^{b_i}$ with $a + \sum_i 2ib_i \le D$; we call D the *degree*. Its dimension is 707, 846 and 1041 for $D = 20$, 21 and 22. A "cap-free optimiser of degree D" is a rational approximation of the top generalised eigenvector of (kJ_0, I_0) on V_D; the exact vectors used are stored in `code/cap6/` (`P_capfree_D20_u.json`, `P_ax45_D21_u.json`, `P_kin48_D22r_u.json`), and every value below is for that stored vector.

Two independent exact evaluators. Given a polynomial P with rational coefficients in some V_D, both programs return $I_T(P\,1_T)$ and $J_T(P\,1_T)$ as exact rationals.

(1) `capped_exact.py` computes the cap-free values with Lemma 1 and subtracts the corrections of Proposition 2, piece by piece.
(2) `capped_check.py` was written separately from the definitions only, without access to the first program. It sums directly over the exact large set, with exponential-moment generating functions, and has no cap-free term (details in [2, §5]).

Both were tested independently: a Monte Carlo comparison at $k = 12$ with caps that remove almost half of I and J; exact agreement of `capped_check.py` with brute-force polytope triangulation and symbolic integration for $k = 2, \dots, 5$; and exact agreement of the two programs on every production input reported as "two programs" below. A value counts as certified when both programs return the same two rationals.

Optimising on the capped support, and ceilings. Both functionals are quadratic forms, $I_T = c^\top G_I c$ and $J_T = c^\top G_J c$, in the coefficients c of P. Running the chain of exact linear maps of `capped_check.py` backwards gives the Gram matrices at about the cost of one evaluation (`capped_gram.py`; Proved here, elementary). We use them in two ways. First, to *find* a good vector (the top generalised eigenvector), which is then rounded and re-evaluated exactly by the evaluators; only that exact value is stated as a certificate. Second, to prove a *ceiling*: if $\sigma G_I - kG_J$ is positive definite, no P in the space reaches the ratio σ. Positive definiteness is certified by recursive Schur complements in ball arithmetic, with the entrywise error of G_J included in the balls. The weights behind G_I and G_J were checked against `capped_check.py` exactly at a uniformly random integer vector, which by the Schwartz–Zippel lemma rules out a discrepancy between the two quadratic forms except with probability below 2^{-31} for each.

4. Results

The support of Stadlmann ($k = 49$, claimed $H_1 \le 240$).

Theorem 3 (Computed; exact, two independent programs)**.** *For the parameters of* [7] *in the table above:*

(1) *for P the cap-free optimiser in V_{20} (707 coefficients),*

$$49J_T(P1_T)/I_T(P1_T) = 4.0120389035\ldots;$$

(2) *for P^* the top eigenvector of $(49G_J, G_I)$ in the same space, rounded to 400 significant bits per coefficient,*

$$49J_T(P^*1_T)/I_T(P^*1_T) = 4.0124198475\ldots.$$

In both cases the two evaluators agree on I_T and J_T as exact rationals. Without caps the same space gives 4.016176, so the caps cost 0.0041 for the truncated optimiser and 0.0038 for the optimised one: optimising on T regains less than a tenth of the loss. Since $T \subset L\mathcal{R}_{49}$, $\sup_F kJ_T/I_T$ is at most the cap-free constant $M_{49}(1.042, 0.982)$, which the dual-weight theorem of [2, §4] (weight parameters $a = 1.0417$, $s_1 = 1.0007$) bounds by 4.14; so $\sup_F kJ_T/I_T \le 4.14$ over *all* F (COMPUTED, interval arithmetic, `dual_eps_cert.certify_box`, 337 pieces). So the margin of the variational step of [7] lies in (0.0124, 0.14].

Corollary 4 (conditional; the analytic input is CLAIMED)**.** *The criterion of* [7, Prop. 1] *holds for $k = 49$ with Stadlmann's support, caps and parameters. With the* LEAN-CHECKED *admissible 49-tuple of diameter 240, $H_1 \le 240$ follows from* [7, Prop. 1] *and his equidistribution theorem, which we have not verified.*

The support of Kintali and of Song–Yue ($k = 48$, claimed $H_1 \le 236$). Both preprints use the same support (the table above, called "D182" in [3]) and, by their descriptions, the same public witness of degree 25, originally constructed for $k = 50$ and evaluated at $k = 48$ without re-optimisation. Song and Yue print it as 1295 integer triples (a, α, c), with

$$P(z) = \sum c\,\left(\tfrac{26}{25} - \textstyle\sum_i z_i\right)^a m_\alpha(z), \qquad P_{\mathrm{SY}}(t) = P\Big(\tfrac{26}{25U_{\mathrm{SY}}}t\Big), \quad U_{\mathrm{SY}} = \tfrac{521}{2000},$$

m_α the monomial symmetric polynomials, and they print the reference integers $I_{\mathrm{ref}} = (98!)^2 I_0^*$ and $J_{\mathrm{ref}} = (99!)^2(26!)^2 J^*$. Here I_0 is the *cap-free* mass on $\{\sum_i t_i < U_{\mathrm{SY}}\}$, J the capped numerator, both in their t-coordinates, and the stars denote integer coordinates $x = 10^4 t$. Their criterion is $48J/I_0 > 2001/2000$ (threshold 1 in their normalisation), which implies the capped criterion because $I_T \le I_0$ (LITERATURE, [6, §4.12, App. A, App. C]). They state that a fresh evaluation reproduces I_{ref}, while fresh regeneration of J_{ref} from the table "remains incomplete".

We extracted the table from the PDF using font sizes to read the exponents in the partitions. All 1295 rows are present, every α is nondecreasing with even parts, and $\max(a + |\alpha|) = 25$, as stated. Since all parts are even, P_{SY}, rewritten in $u = 4t$, is exactly a polynomial in $L - P_1$ and the even power sums $P_2, \ldots, P_{24}$ (1311 terms); the conversion was checked exactly at random rational points.

Theorem 5 (COMPUTED, exact rational arithmetic; items (1) and (2) each by two independent programs)**.** *For the witness P_{SY} of* [6] *and the support above, with I_0 and J_T computed in the t-coordinates of* [6]*:*

(1) $(98!)^2\,10^{192}\,5^{150}\,521^{50}\,I_0 = I_{\mathrm{ref}}$ *exactly (an integer with 655 digits);*

(2) $(99!)^2(26!)^2\,10^{196}\,5^{150}\,521^{50}\,J_T = J_{\mathrm{ref}}$ *exactly (an integer with 715 digits), where J_T is the capped numerator;*

(3) *hence, in the normalisation of* [6] *(threshold 1), $48J_T/I_0 = 1.0006649215907\ldots > 2001/2000$. In the Polymath scaling (threshold 4) the same ratio is $4.00265968636\ldots$; with the capped denominator $48J_T/I_T = 4.0030988047\ldots$, and without caps $48J_0/I_0 = 4.0036348167\ldots$.*

The factors $10^{192}5^{150}521^{50}$ and $10^{196}5^{150}521^{50}$ come from the change to integer coordinates when the polynomial is scaled by 65125^{25} (our reading of [6, Eq. (71)]; $65125 = 25 \cdot 2605$),

so items (1) and (2) are equalities of integers, not numerical agreement. Item (1) was computed by `songyue_I0.py`, written for this purpose, which works in the monomial basis of the table, integrating each product $m_\alpha m_\beta$ by a sum over partial matchings of parts, and does not use the power-sum conversion; `songyue_J0.py` computes J_0 the same way. Item (2) was computed by `capped_check.py` on the converted polynomial (56 minutes on four cores). Kintali [3] prints, in the normalisation with threshold 1, $48J_T/I_T = 1.000774701187468\ldots$ and $48J_0/I_0 = 1.000908704175987\ldots$, and the ratios $I_T/I_0 = 0.99989030538\ldots$ and $J_T/J_0 = 0.99975643873\ldots$; ours agree with every digit he prints. For the capped ratio he prints 33 digits; four times his value and our exact value are

$$4 \times 1.000774701187468444763475111497864 = 4.003098804749873779053900445991456,$$

$$48J_T/I_T = 4.003098804749873779053900445991457 75\ldots.$$

Second program (version 0.3). The second evaluator, `capped_exact.py` (cap-free value minus 126 inclusion–exclusion correction pieces, each an exact rational; about 32 core-hours), gives the same J_T as `capped_check.py` as an exact rational, so item (2) holds by two independent programs; its cap-free values equal I_0 and J_0 of the monomial-basis programs `songyue_I0.py` and `songyue_J0.py` exactly, after the change of coordinates, so item (1) does too. The J_T comparison is recorded in `code/cap6/compare2_sy236.json` (script `code/songyue_compare2.py`); the cap-free comparison was made when the run finished and is recorded in the project log (`log/entries.json`, Session 12), not in that file. The capped denominator I_T, which enters only the ratio $48J_T/I_T$ and not the criterion of [6], rests on `capped_check.py` alone. Proposition 6 below is also confirmed by both programs (version 0.4).

Proposition 6 (Computed, exact rational arithmetic, two independent programs)**.** *For our own cap-free optimiser in V_{22} for this (L,τ) at $k = 48$ (1041 coefficients, rounded to 400 significant bits), $48J_T(P1_T)/I_T(P1_T) = 4.0060814208\ldots > 4$.*

`capped_check.py` and `capped_exact.py` (176 correction pieces, 15.5 core-hours) give the same I_T and J_T as exact rationals (`code/cap6/compare_kin48_D22r.json`). Here the caps cost only 0.00033 (the cap-free value is 4.0064115). That our value exceeds the witness's 4.0031 is expected, since the witness was optimised for $k = 50$ and transported, while ours is optimised for $k = 48$.

Corollary 7 (conditional; the analytic input is Claimed)**.** *For the published witness, the variational criteria of both [3] ($48J_T/I_T > 4$) and [6] ($48J_T/I_0 > 2001/2000$) hold at $k = 48$. For our independent test function only the capped criterion $48J_T/I_T > 4$ was computed, and it holds (two programs). With the* Lean-checked *admissible 48-tuple of diameter 236, $H_1 \le 236$ follows from their equidistribution input, which we have not verified.*

The draft of Charton et al. ($k = 45$, claimed $H_1 \le 212$).

Proposition 8 (Computed; exact, two independent programs)**.** *With the caps of [1, Table 3] and P our cap-free optimiser in V_{21} (846 coefficients; V_{21} is the space $\mathcal{B}_{21}$ of [1, Thm. 11.1], which is defined there as the span of $(L - P_1)^a P_\lambda$ with λ having even parts and $a + |\lambda| \le 21$;* Literature*), $45J_T(P1_T)/I_T(P1_T) = 4.0027351128\ldots > 4$.*

The authors report 4.00438 with their own optimised function (Claimed); our value is lower, as expected for a function optimised without caps, and already above 4. The draft, however, also states caps in its Lemma 9.1, namely $C_1 = C_2 = 31/50$ and $C_m = 17/25$ for $m \ge 3$, which are smaller than those of Table 3 for every $m \ge 1$. With these the conclusion changes.

Proposition 9 (Computed, ball arithmetic; Gram weights from one program, checked at a random point)**.** *Let T' be the support with the caps of Lemma 9.1 of [1] and $V = V_{21}$. Then*

$3.99\,G_I - 45\,G_J$ is positive definite on V, so

$$\frac{45 J_{T'}(P1_{T'})}{I_{T'}(P1_{T'})} < 3.99 \qquad \textit{for every nonzero } P \in V.$$

The weaker statement with 4 in place of 3.99 was certified twice, with halved blocks (the project log records 4800 bits; the run log does not record the precision) and at 6500 bits (blocks split $1:2$); the statement with 3.99 was certified once, at 6500 bits. Both runs use the same exact weights, computed by running `capped_check.py` backwards (`capped_gram.py`) and checked against `capped_check.py` exactly at one random integer point (Schwartz–Zippel). The second program, `capped_exact.py`, has not yet finished evaluating that point, so this proposition rests on one program; and the saved weights themselves are not in the repository (they are regenerated by `capped_gram.py`). The best P in V gives about 3.98555 (floating point, from the Gram matrices). Since V is the authors' own space, the caps of Lemma 9.1 cannot give their 4.00438, which is consistent with the caps of Table 3. We report this as a located inconsistency for the authors to resolve, not as an error in their result.

5. Remarks

Remark 10 (on [7, Lemma 5]; Literature reading, not a result). Song and Yue [6, Remark 4.23] say that one step in the proof of [7, Lemma 5] lacks a hypothesis. In arXiv v1, Lemma 5 (a Type I estimate) takes $f = \alpha \star \beta$ with α an arbitrary coefficient sequence and β smooth. For γ in a short range above $1/2$ the proof applies Lemma 3 (a Type II estimate) "with the roles of α and β swapped"; Lemma 3 requires its shorter factor to satisfy a Siegel–Walfisz condition, and after the swap that factor is α, for which neither Lemma 5 nor the case of [7, Definition 9] where it is used assumes this. So the step reads as [6] describe it. A natural repair is to check that the sequences α arising in the decomposition have the property; we have not done so, and we do not know whether the conclusion is affected. Theorem 3 concerns only the variational step and is unaffected; the *bound* 240 depends on this input.

The question may also bear on the 236 of [3]: Kintali proves no new estimate and takes [7, Prop. 3], with its Type I, IIa, IIb and III branches and the Type IIc range made empty, as his analytic input (Literature). Whether his argument is affected depends on whether those branches pass through the step in question, which we have not traced. Song and Yue prove their own estimate [6, Thm. 4.18], via a Heath–Brown decomposition and Type IIa, IIb and III estimates [6, §§4.2–4.7]; we have not checked whether it uses an argument of the same kind. Charton et al. [1] list the same five Type I, II and III estimates as [7], Lemma 5 among them, so the same question applies to their draft. Theorem 5 and Proposition 8 are unaffected either way.

Remark 11 (what would change the picture). The variational steps examined here have margins of 0.0027 to 0.012 above the threshold 4; against Song and Yue's own criterion $48J/I_0 > 2001/2000$ the witness clears by only $1.6 \cdot 10^{-4}$ (that is, $6.6 \cdot 10^{-4}$ in the Polymath scaling). They are therefore robust to rounding but not to a change of support: Proposition 9 shows that a modest change in the caps moves the best ratio in V_{21} below 4. Any revision of the analytic input that shrinks the admissible caps should be re-checked against the variational step, which our programs can do in hours.

6. Reproducibility

All inputs, certificates and logs are in the repository, under `code/` and `code/cap6/`. The evaluators are `code/capped_exact.py` and `code/capped_check.py`; the Gram matrices and ceilings come from `code/capped_gram.py`; the Song–Yue witness and reference integers, as extracted,

are `code/cap6/songyue236_witness.json` and `code/cap6/songyue236_appC.json`, with the conversion and comparison scripts `code/songyue_*.py`. Running times above are for four cores and 16 GB of memory. The admissible tuples of diameters 212, 236, 240 and 246 are checked by the Lean 4 kernel (core Lean, no Mathlib, no `sorry`; axioms `propext` and `Quot.sound` only), as are the counting lemmas used in Proposition 2. The companion paper [2] contains the full derivations and the history of each computation: the dual-weight ceilings in its §4, and the capped supports, the two evaluators and the Gram matrices in its §5 (section numbers of its version 0.12).

Changes

0.4: (14 November 2026) The second exact program confirms Proposition 6 (our own $k = 48$ test function): same I_T and J_T as exact rationals. The one-program qualifiers for it are removed from the abstract, the table, the proposition and the corollary.

0.3: (13 November 2026) The second exact program confirms the capped numerator of the witness of [6] (Theorem 5); the one-program qualifiers for it are removed from the abstract, the table, the theorem and the corollary.

0.2: (13 November 2026) Revised after an internal referee report (in the directory `referee/` of the repository). The normalisations are now stated (new paragraph in §2) and the 236 ratios in Theorem 5 and the results table are given in one scaling; the one-program status of the 236 capped numerators is marked in the table, the theorem and the corollary, and the corollary no longer claims the criterion of [6] for our own test function; the cap-free agreement of the second evaluator is attributed correctly; the 4.14 ceiling has its provenance; "certified twice" is corrected for the 3.99 ceiling, and the statements after Proposition 9 are restricted to V_{21}, with a conjecture about the drafting of [1] removed; Proposition 2 sets $C_0 = 0$ and assumes the cap condition for $m \geq 1$ only, with the case $m(u') = 0$ added to the proof; the Lean attributions name the theorems and what they bound; the test spaces V_D are defined; overloaded symbols are renamed ($\mathcal{L}$, θ, U_{SY}, I_0, J_0); "$m \geq 4$" is corrected to "$m \geq 1$"; truncated digits are corrected; Remark 10 now discusses which 236 argument may depend on the step it describes.

0.1: (13 November 2026) First draft.

References

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[2] J. Kim, *Toward the twin prime conjecture: certificates, limits of the sieve, and a working programme*, living paper, 2026. https://jerryckim16.github.io/twin-primes/

[3] S. Kintali, *Improved bounded gaps between primes*, preprint, 1 September 2026. https://shivakintali.github.io/papers/Prime-Gaps.pdf

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[5] D. H. J. Polymath, *Variants of the Selberg sieve, and bounded intervals containing many primes*, Res. Math. Sci. 1 (2014).

[6] Z. Song, S. Yue, *Bounded gaps between primes: an upper bound of 236*, IACR ePrint 2026/1893, version of 2 September 2026.

[7] J. Stadlmann, *Bounded gaps between primes*, arXiv:2608.31126, 31 August 2026.